

1002/01994

Maths

Page - I

1.  $\rightarrow (3) 3 \checkmark$

2.  $\rightarrow (2) \sin 60^\circ \checkmark$

3.  $\rightarrow (4) 100 \checkmark$

4.  $\rightarrow (1) 10 \text{ cm} \checkmark$

5.  $\rightarrow (1) a = 0 \checkmark$

6.  $\rightarrow (4) \text{ Four decimal places} \checkmark$

7.  $\rightarrow (3) 3 \text{ and } 1 \checkmark$

8.  $\rightarrow (2) (0, -1) \checkmark$

9.  $\rightarrow (1) \text{ Unique sol } (4) \text{ No solution} \checkmark$

10.  $\rightarrow (1) -\frac{9}{2} \checkmark$

11.  $\rightarrow (1) K \neq -3 \checkmark$

12.  $\rightarrow (1) 8 \checkmark$

13.  $\rightarrow (1) \text{ Similar} \checkmark$

14.  $\rightarrow (1) x^3 y^4 z^4 \checkmark$

15.  $\rightarrow (3) \frac{x^2}{2} - \frac{x}{2} - 6 \checkmark$

Page - 2

16.  $\rightarrow$  (3) intersecting or coincident  $\checkmark$

17.  $\rightarrow$  (1) Parallel  $\checkmark$

18.  $\rightarrow$  (4) 16 : 81  $\checkmark$

19.  $\rightarrow$  (3) Assertion (A) is true but Reason (R) is False  $\checkmark$

20.  $\rightarrow$  (1) Both A and R are true and R is the correct explanation of A  $\checkmark$

### Section - B

21.  $\rightarrow$  Given - a number in the form,

$$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$$

$$144 \times 5 + 5$$

$$5(144 + 1)$$

$$(5)(145)$$

$$(5)(5)(29)(1)$$

$\therefore$  The given no. is a composite number as it has more than two factors.

22.  $\rightarrow$  Given - Pair of linear equations -

$$4x + 6y = 7; \quad kx + 18y = 21$$

Solution - Condition for infinitely many solutions

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}, \quad \text{or}$$

page-3

$$\frac{4}{K} = \frac{1}{3} = \frac{7}{21}$$

$$\frac{4}{K} = \frac{1}{3}$$

$$4 \times 3 = K$$

$$12 = K$$

$\therefore$  The value of  $K$  will be 12 for linear equation pair with coincident lines.

3.  $\rightarrow$  Given - point  $(-6, 8)$ , Finding its distance from origin,

Solution - Using the formula -

$$= \sqrt{x^2 + y^2}$$

$$= \sqrt{(-6)^2 + (8)^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100} = 10 \text{ units}$$

$\therefore$  The distance of the point from origin is 10 units.

4.  $\rightarrow$  Given -  $\sin 3A = \cos(A - 10^\circ)$  finding  $A$  -

Solution -

$$\sin 3A = \cos(A - 10^\circ)$$

$$\cos(90^\circ - 3A) = \cos(A - 10^\circ) \dots [\sin A = \cos(90^\circ - A)]$$

$$90^\circ - 3A = A - 10^\circ$$

$$90^\circ + 10^\circ = A + 3A$$

$$100^\circ = 4A$$

$$A = \frac{100^\circ}{4} = 25^\circ$$

$$\therefore A = 25^\circ$$



25 → Given - Fig. with  $\triangle AOB$  and  $\triangle DOC$ ,  $AB \parallel DC$

To prove -  $\triangle AOB \sim \triangle DOC$ ,

Solution - In  $\triangle AOB$  and  $\triangle DOC$ ,

$$\angle AOB = \angle DOC \quad \dots [\text{Vertically opposite angles}]$$

$$\angle ABO = \angle DCO \quad \dots [\text{Alternate interior angles}]$$

$$\angle BAO = \angle CDO \quad \dots [\text{Alternate interior angles}]$$

$$\therefore \triangle AOB \sim \triangle DOC \quad \dots [\text{By AA similarity}]$$

$\therefore$  Hence, proved

26 →

Section - C

26 → Given - To prove -  $\frac{\cos(90^\circ - \theta)}{1 + \sin(90^\circ - \theta)} + \frac{1 + \sin(90^\circ - \theta)}{\cos(90^\circ - \theta)} = 2 \operatorname{cosec} \theta$

Solution - Using  $\cos(90^\circ - \theta) = \sin \theta$  and Solving LHS  
 $\sin(90^\circ - \theta) = \cos \theta$

$$\frac{\cos(90^\circ - \theta)}{1 + \sin(90^\circ - \theta)} + \frac{1 + \sin(90^\circ - \theta)}{\cos(90^\circ - \theta)} = 2 \operatorname{cosec} \theta$$

$$= \frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} \quad \dots [\cos(90^\circ - \theta) = \sin \theta, \sin(90^\circ - \theta) = \cos \theta]$$

$$= \frac{\sin \theta \times \sin \theta}{1 + \cos \theta \times \sin \theta} + \frac{1 + \cos \theta}{\sin \theta \times 1 + \cos \theta}$$

$$= \frac{\sin^2 \theta}{\sin \theta + \sin \cos \theta} + \frac{(1 + \cos \theta)^2}{\sin \theta + \sin \cos \theta}$$

$$= \frac{\sin^2 \theta + 1 + \cos^2 \theta + 2 \cos \theta}{\sin \theta + \sin \cos \theta} \quad \dots [\text{Using } (a+b)^2 = a^2 + b^2 + 2ab]$$

$$= \frac{\sin^2 \theta + \cos^2 \theta + 1 + 2 \cos \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{1 + 1 + 2 \cos \theta}{\sin \theta (1 + \cos \theta)} \quad \dots [\sin^2 \theta + \cos^2 \theta = 1]$$

$$\frac{2 + 2\cos\theta}{\sin\theta(1+\cos\theta)}$$

$$\frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)},$$

$$= \frac{2}{\sin\theta}$$

$$2\operatorname{cosec}\theta = 2\operatorname{cosec}\theta \dots\dots \left[ \frac{1}{\sin\theta} = \operatorname{cosec}\theta \right]$$

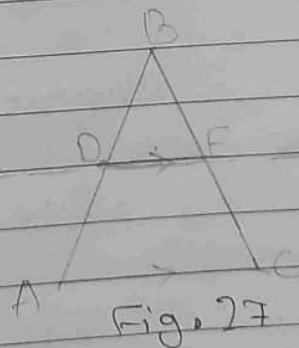
$$\therefore \text{LHS} = \text{RHS}$$

$\therefore$  Hence proved

27.  $\rightarrow$  Basic proportionality theorem or Thales theorem states that if a line parallel to one of the sides of the triangle when passes through the other two sides, then the other two sides will be in the same ratio.

To prove -  $\frac{AD}{BD} = \frac{BE}{EC}$

Solution - Proving  $\triangle ABC \sim \triangle DBE$ ,



In  $\triangle ABC$  and  $\triangle DBE$

$$\angle BAC = \angle BDE$$

$$\angle BAC = \angle BDE \dots \text{[Corresponding angles]}$$

$$\angle ABC = \angle DBE \dots \text{[Common]}$$

$$\therefore \triangle ABC \sim \triangle DBE \dots \text{[By AA similarity]}$$

By similarity

$$\frac{AB}{DB} = \frac{CB}{EC}$$



page-6

$$\frac{AD+DB}{DB} = \frac{CE+BE}{CE}$$

$$\frac{AD}{BD} + \frac{BD'}{BD} = \frac{CE'}{CE} + \frac{BE}{CE}$$

$$\therefore \frac{AD}{BD} = \frac{BE}{EC}$$

$\therefore$  Hence proved.

28  $\rightarrow$  Given - Book when rented has fixed charge for two days and an extra charge for each day after that,

Latika paid ₹22 for 6 days while

Anand paid ₹16 for 4 days

Solution - Consider the fixed charge as  $x$  and extra charge as  $y$ , then

Charge for Latika -

$$₹22 = x + \cancel{4y} (\text{total days} - 2)y$$

$$₹22 = x + (6-2)y = x + 4y \dots \text{eq (i)}$$

Charge for Anand -

$$₹16 = x + (\text{total days} - 2)y$$

$$= x + (4-2)y$$

$$₹16 = x + 2y \dots \text{eq (ii)}$$

Subtracting eq (ii) from eq (i),

$$x + 4y - (x + 2y) = 22 - 16 \dots [\text{In } ₹]$$

$$x + 4y - x - 2y = 6 \dots [\text{In } ₹]$$

$$4y - 2y = 6$$

$$2y = 6$$

$$y = \frac{6}{2} =$$

$$y = ₹3$$

Substituting value of  $y$  in eq (ii),

$$₹16 = x + 2(₹3)$$

$$₹16 = x + ₹6$$

$$₹16 - ₹6 = x$$

$$₹10 = x$$

∴ The fixed charge for first two days is ₹10 and additional charges for next days is ₹3 per day.

29. → Given - Points  $A(-5, 7)$ ,  $B(-4, 5)$ ,  $C(-1, -6)$  and  $D(4, 5)$  forms quadrilateral.

To find - Area enclosed by quadrilateral ABCD,

Solution - Finding AC and using area of quad

Solution - Dividing the quadrilateral into  $\triangle ABC$  and  $\triangle ADC$  and finding their areas by

$$\text{Area of triangle} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Page-8

In  $\triangle ABC$ ,  $x_1 = -5$ ,  $x_2 = -4$ ,  $x_3 = -1$   
 $y_1 = 7$ ,  $y_2 = -5$ ,  $y_3 = -6$

Putting the values in the formula,

$$= \frac{1}{2} |-5[-5+6] + (-4)(-6-7) + (-1)[7-(-5)]|$$

$$= \frac{1}{2} |-5(-1) + (-4)(-13) + (-1)(12)|$$

$$= \frac{1}{2} |-5 + 52 - 12|$$

$$= \frac{1}{2} |-5 + 40|$$

$$= \frac{1}{2} \times 35$$

$$= \frac{35}{2} \text{ square units}$$

In  $\triangle ADC$ ,  $x_1 = -5$ ,  $x_2 = 4$ ,  $x_3 = -1$   
 $y_1 = 7$ ,  $y_2 = 5$ ,  $y_3 = -6$

Area of  $\triangle ADC$  is

$$= \frac{1}{2} |-5(5+6) + 4(-6-7) - 1(7-5)|$$

$$= \frac{1}{2} |-5(11) + 4(-13) - 1(2)|$$

$$= \frac{1}{2} |-55 - 52 - 2|$$

$$= \frac{1}{2} |-109|$$

$$= \frac{109}{2} \text{ square units}$$



$$\begin{aligned}
 &\text{Area of quadrilateral ABCD} - \\
 &= \text{Area of } \triangle ABC + \text{Area of } \triangle ADC \\
 &= \frac{35}{2} + \frac{109}{2} \\
 &= \frac{144}{2} \\
 &= 72 \text{ Sq. units}
 \end{aligned}$$

$\therefore$  Area of quadrilateral ABCD is 72 sq. units

30.  $\rightarrow$  Given - Two numbers 861 and 1353  
To Find - Their HCF using Euclid's algorithm

Solution - Since  $1353 > 861$  we will apply lemma on 135 and 861

$$1353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

$$492 = 369 \times 1 + 123$$

$$369 = 123 \times 3 + 0$$

Since remainder is zero when 123 came, 123 is the HCF

$\therefore$  HCF of 1353 and 861 is 123,

31.  $\rightarrow$  Given - Points  $A(5, -2)$  and  $B(-3, 2)$   
 To find - Point on  $x$ -axis equidistant from both points.

Solution - Let the point on  $x$ -axis be  $P(x, 0)$   
 Using distance formula,  $\sqrt{(x-x_1)^2 + (y-y_1)^2}$

$$PA = PB$$

$$\sqrt{(x+5)^2 + (0+2)^2} = \sqrt{(x-3)^2 + 0^2}$$

$$\sqrt{(x+5)^2 + (0+2)^2} = \sqrt{(x-3)^2 + (0-2)^2}$$

Squaring both sides, we get

$$(x+5)^2 + (2)^2 = (x-3)^2 + (-2)^2$$

$$x^2 + 25 - 10x + 4 = x^2 + 9 - 6x + 4$$

$$25 - 9 = 10x - 6x$$

$$16 = 4x$$

$$x = \frac{16}{4} = 4$$

$$x = 4$$

$\therefore$  The point on the  $x$ -axis which is equidistant from points  $A$  and  $B$  is  $P(4, 0)$

32.  $\rightarrow$  Given -  $\tan A + \sin A = m$ ;  $\tan A - \sin A = n$ ,  
 To prove -  $m^2 - n^2 = 4 \sqrt{mn}$

Solution - First solving LHS,

Page - 11

$$\begin{aligned}
 \text{LHS} &= (\tan A + \sin A)^2 + (\tan A - \sin A)^2 \\
 &= \tan^2 A + \sin^2 A + 2 \tan A \sin A - (\tan^2 A + \sin^2 A - 2 \tan A \sin A) \\
 &= \cancel{\tan^2 A} + \cancel{\sin^2 A} + 2 \tan A \sin A - \cancel{\tan^2 A} - \cancel{\sin^2 A} + 2 \tan A \sin A \\
 &= 4 \tan A \sin A
 \end{aligned}$$

Now solving RHS, we get,

$$\begin{aligned}
 &= 4 \sqrt{mn} \\
 &= 4 \sqrt{(\tan A + \sin A)(\tan A - \sin A)} \\
 &= 4 \sqrt{\tan^2 A - \sin^2 A} \\
 &= 4 \sqrt{\frac{\sin^2 A}{\cos^2 A} - \sin^2 A} \quad \dots \quad \left( \tan^2 A = \frac{\sin^2 A}{\cos^2 A} \right) \\
 &= 4 \sqrt{\frac{\sin^2 A - \cos^2 A \sin^2 A}{\cos^2 A}} \\
 &= 4 \sqrt{\frac{\sin^2 A (1 - \cos^2 A)}{\cos^2 A}} \\
 &= 4 \sqrt{\frac{\sin^2 A \times \sin^2 A}{\cos^2 A}} \quad \dots \quad [\sin^2 A = 1 - \cos^2 A] \\
 &= \frac{4 \sin A \times \sin A}{\cos A} \\
 &= 4 \tan A \sin A
 \end{aligned}$$

 $\therefore \text{LHS} = \text{RHS}$ , Hence proved



34.  $\rightarrow$  Given -  $p(x) = 2x^3 - x^2 - 4x + 2$  with two zeroes -  $\sqrt{2}$  and  $-\sqrt{2}$

To Find - third zero of  $p(x)$

Solution - In  $p(x)$ ,  $a=2$ ,  $b=-1$ ,  $c=-4$ ,  $d=2$

$$\text{Using, } \alpha + \beta + \gamma = -\frac{b}{a}$$

Substituting values of  $\alpha$ ,  $\beta$  and  $b$  and  $a$  in the formula

$$(\sqrt{2}) + (-\sqrt{2}) + \gamma = -\frac{(-1)}{2}$$

$$0 + \gamma = \frac{1}{2}$$

$$\gamma = \frac{1}{2}$$

$\therefore$  The third root of the polynomial is  $\frac{1}{2}$ .

35 → Given - line segment with endpoints  $(4, -3)$  and  $(8, 5)$  divided internally in ratio  $3:1$ .

To find - Coordinates of the point of division

Solution - Consider the point as  $P$ ,  
Using the Section Formula to find,

$$P = \left( \frac{x_2 m_1 + x_1 m_2}{m_1 + m_2}, \frac{y_2 m_1 + y_1 m_2}{m_1 + m_2} \right)$$

$$\text{Here } x_1 = 4, x_2 = 8, m_1 = 3, m_2 = 1$$

$$y_1 = -3, y_2 = 5$$

Putting values,

$$P = \left( \frac{8(3) + 4(1)}{3+1}, \frac{5(3) + (-3)(1)}{3+1} \right)$$

$$P = \left( \frac{24+4}{4}, \frac{15-3}{4} \right)$$

$$P = \left( \frac{28}{4}, \frac{12}{4} \right)$$

$$P = (7, 3)$$

∴ The point  $P(7, 3)$  divide the line joining  $(4, -3)$  and  $(8, 5)$  into  $3:1$

36.  $\Rightarrow$  i)  $\rightarrow$  (2) (0, 12)  $\checkmark$

ii)  $\rightarrow$  (1) (3, 5)  $\checkmark$

iii)  $\rightarrow$  (3) 5m  $\checkmark$

37.  $\Rightarrow$  i)  $\rightarrow$  (3) Parabola  $\checkmark$

ii)  $\rightarrow$  (3) 0  $\checkmark$

iii)  $\rightarrow$  (i) 3, 2  $\checkmark$

38.  $\Rightarrow$  i) (3)  $24 \text{ cm}^2$

ii)  $\rightarrow$

iii)  $\rightarrow$  2)  $4\frac{9}{8}$   $\checkmark$

or

iii)  $\rightarrow$  (4) converse of pythagoras theorem  $\checkmark$



33 → Given linear equations  $x - y + 1 = 0$ ,  $3x + 2y - 12 = 0$

In  $x - y + 1 = 0$ , when

|     |   |   |   |
|-----|---|---|---|
| $x$ | 1 | 2 | 5 |
| $y$ | 2 | 3 | 6 |

In  $3x + 2y - 12 = 0$

|     |     |   |
|-----|-----|---|
| $x$ | 1   | 2 |
| $y$ | 4.5 | 3 |

33 ->

$$3x + 2y - 12 = 0$$

$$x - y + 1 = 0$$

(4, 5)

(5, 6)

(2, 3)

(1, 2)

